COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 Data Structures And Algorithms	**SECTION:**	CPE21S2
DATE PERFORMED:	09/16/26	**DATE SUBMITTED:**	09/1626
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HOA 7.1 Sorting Algorithms

1. PROCEDURE OUTPUT

main.cpp

```cpp
#include <iostream>
#include "BinaryTree.h"

int main()
{
    // 1. Creating the root node:
    BinaryTree<char> *root = new BinaryTree<char>('A');
    std::cout << "Root created: " << root->getRootVal() << std::endl;

    // 2. Inserting left and right children of root:
    root->insertLeftChild('B');  // left child of A
    root->insertRightChild('C'); // right child of A
    std::cout << "Root left child: " << root->getLeftChild()->getRootVal() << std::endl;
    std::cout << "Root right child: " << root->getRightChild()->getRootVal() << std::endl;

    // 3. Inserting children of node B:
    root->getLeftChild()->insertLeftChild('D');  // left child of B
    root->getLeftChild()->insertRightChild('E'); // right child of B
    std::cout << "Node B left child: " << root->getLeftChild()->getLeftChild()->getRootVal() << std::endl;
    std::cout << "Node B right child: " << root->getLeftChild()->getRightChild()->getRootVal() << std::endl

    // 4. Inserting children of node C:
    root->getRightChild()->insertLeftChild('F');  // left child of C
    root->getRightChild()->insertRightChild('G'); // right child of C
    std::cout << "Node C left child: " << root->getRightChild()->getLeftChild()->getRootVal() << std::endl;
    std::cout << "Node C right child: " << root->getRightChild()->getRightChild()->getRootVal() << std::end

    // 5. Testing setRootVal (changing a node's value):
    root->getLeftChild()->getRightChild()->setRootVal('N'); // change E → N
    std::cout << "Node B right child after setRootVal: "
              << root->getLeftChild()->getRightChild()->getRootVal() << std::endl;

    // 6. Testing getRootVal again after modification:
    std::cout << "Root value still: " << root->getRootVal() << std::endl;

    return 0;
}
```

```cpp
#ifndef BINARYTREE_H
#define BINARYTREE_H

template <typename T>
class BinaryTree
{
private:
    T key;
    BinaryTree<T> *leftchild;
    BinaryTree<T> *rightchild;

public:
    BinaryTree(T rootObj);

    void insertLeftChild(T rootobj);
    void insertRightChild(T rootobj);

    BinaryTree<T> *getLeftChild();
    BinaryTree<T> *getRightChild();

    T getRootVal();
    void setRootVal(T obj);
};

template <typename T>
BinaryTree<T>::BinaryTree(T rootObj)
{
    key = rootObj;
    leftchild = nullptr;
    rightchild = nullptr;
}

template <typename T>
void BinaryTree<T>::insertLeftChild(T newNode)
{
    if (leftchild == nullptr)
    {
        leftchild = new BinaryTree<T>(newNode);
    }
    else
    {
        BinaryTree<T> *newChild = new BinaryTree<T>(newNode);
        newChild->leftchild = leftchild;
        leftchild = newChild;
    }
}

template <typename T>
void BinaryTree<T>::insertRightChild(T newNode)
{
    if (rightchild == nullptr)
    {
        rightchild = new BinaryTree<T>(newNode);
    }
    else
    {
        BinaryTree<T> *newChild = new BinaryTree<T>(newNode);
```

```
    }
}

template <typename T>
BinaryTree<T> *BinaryTree<T>::getLeftChild()
{
    return leftchild;
}

template <typename T>
BinaryTree<T> *BinaryTree<T>::getRightChild()
{
    return rightchild;
}

template <typename T>
T BinaryTree<T>::getRootVal()
{
    return key;
}

template <typename T>
void BinaryTree<T>::setRootVal(T obj)
{
    key = obj;
}

#endif

```

```
template <typename T>
bool BinaryTree<T>::isLeaf()
{
    return leftchild == nullptr && rightchild == nullptr;
}
```

```
Is root A a leaf? No
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

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Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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